

Improving the Opening Range Breakout Strategy on SPY

Simple, Economically Motivated Enhancements to a Classic Intraday Trading Strategy



CONCRETUM RESEARCH

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The persistence of **Opening Range Breakout (ORB)** strategies as a viable source of compelling risk-adjusted returns remains an open debate. In this article, we conduct a comprehensive **empirical analysis of a classic 5-minute ORB framework applied to SPY over the period January 2008 to December 2025.**

Our results indicate that, in its **plain-vanilla form**, the **strategy fails to generate meaningful net-of-fee returns: commissions and market impact erode** the gross **edge** almost entirely, and participation on virtually every trading session leaves the strategy fully exposed to cumulative trading costs.

Rather than abandoning the ORB concept altogether, however, we ask a different question:

Can a handful of simple, economically motivated improvements transform an otherwise mediocre strategy into an attractive systematic trading framework?

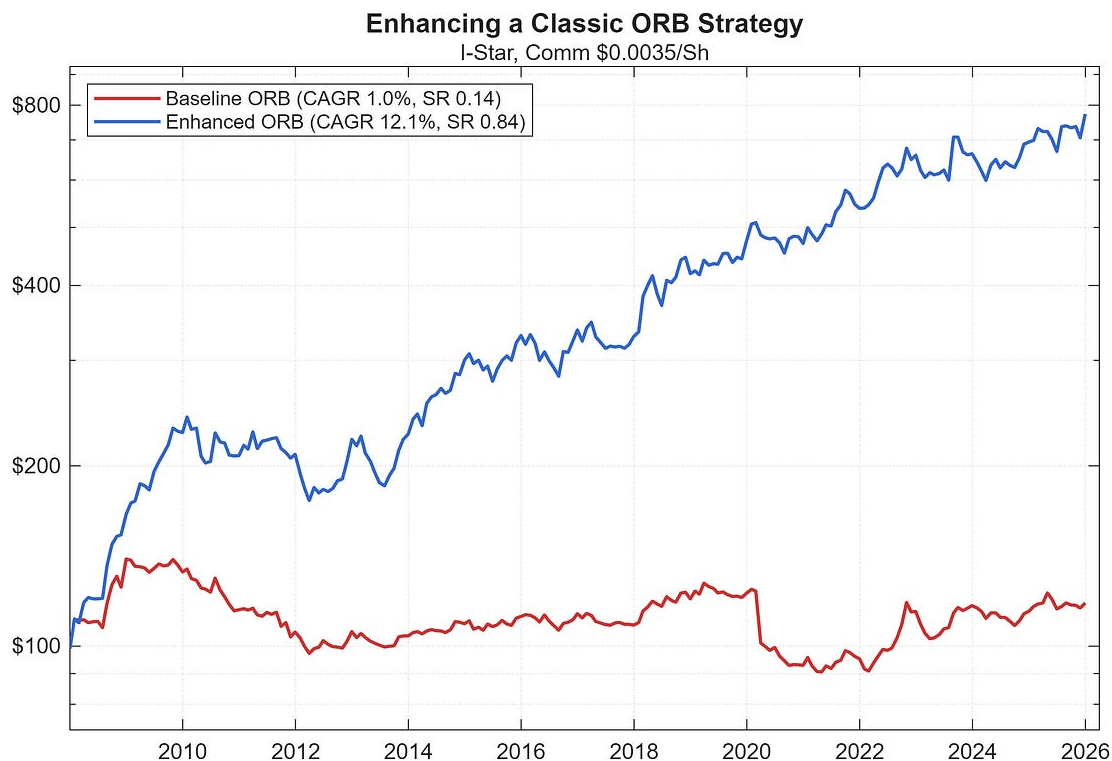
Specifically, we investigate whether a combination of **intuitive trade-selection filters**, improved **trade management** and **dynamic position-sizing** techniques **can** materially **improve** both **returns** and **risk-adjusted performance** without altering the core ORB philosophy or relying on extensive parameter optimization.

Our findings suggest that the answer is yes!

As illustrated in Figure 1, these enhancements increase the strategy's **CAGR from 1.0% to 12.1%**, improve its **Sharpe ratio from 0.14 to 0.84**, and raise **annualized alpha** from an economically modest and **statistically insignificant 1.75% to a highly significant 12.80%**, all while remaining robust under conservative assumptions for commissions and market impact.

The edge lies not in inventing a new ORB signal, but in learning how to trade the existing one more intelligently.

In the remainder of this article, we explain each enhancement individually, discuss the economic intuition behind it, and quantify its incremental contribution to the strategy's overall performance.



Some Background

Among the systematic approaches that have influenced the landscape of intraday trading, **few have proven as enduringly popular as the n -minute opening range breakout (ORB)**. The idea was first formalized by Toby Crabel in *Day Trading with Short Term Price Patterns and Opening Range Breakout*, a foundational work that introduced the use of early-session price action to infer directional conviction.

ORB strategies later gained widespread adoption among practitioners through influential publications such as Connors and Raschke's *Street Smarts*. Over time, they have become one of the most widely studied and replicated frameworks in short-term trading.

Our own research has repeatedly drawn on this logic. Across a series of prior studies, including *Can Day Trading Really Be Profitable?*, *A Profitable Day Trading Strategy for the U.S. Equity Market*, and *Volume Weighted Average Price (VWAP): The Holy Grail for Day Trading Systems*, we have employed the 5-minute ORB as a core directional signal for intraday trend-following strategies, consistently finding it a valuable building block.

Base ORB Strategy

We begin by **establishing the baseline specification of the ORB strategy**, against which all subsequent enhancements will be measured. The implementation is intentionally kept as simple as possible: the goal at this stage is to assess the raw merit of the ORB signal in isolation, before any refinements are applied.

As in our previous work, we focus on the 5-minute opening range breakout (ORB) variant applied to SPY, the largest ETF tracking the S&P 500 Index. At the close of the fifth minute of the regular trading session (RTH), the opening range high (ORH) and opening range low (ORL) are defined as the highest and lowest prices recorded during that initial window: these two levels mark the opening range and will serve as the

directional triggers for the strategy (see Figure 1 for two illustrative examples).



Figure 1: Two illustrative 5-minute ORB entries on SPY

Both entry-exit and position-sizing criteria (summarized below) are straightforward:

Base ORB Strategy	
Instrument	SPY (S&P 500 ETF)
Entry	- Long via stop order at ORH; Short via stop order at ORL - Only one trade per session (first breakout direction)
Exit	- Liquidation upon 5-min close crossing opposite band (market order) - End-of-day via market-on-close (MOC) order
Sizing	Full notional exposure (100% of capital)

A key implementation choice concerns the frequency at which exit conditions are evaluated. Rather than monitoring price at the 1-minute level, we check exit triggers exclusively at 5-minute bar closes. This prevents transient price spikes from prematurely terminating otherwise valid positions, a design choice consistent with our prior work *Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)*, where

both entry- and exit-signal evaluation was performed at 30-minute timestamps.

This choice also reflects the broader principle, well established in quantitative finance, that sub-5-minute price data are prone to microstructure noise and are generally unreliable for signal generation. Additionally, we impose the constraint that a position cannot be closed on the same 1-minute bar on which its entry was originally triggered.¹

¹ Over the sample period (January 2008 - December 2025), there are 24 sessions in which a single 1-minute bar simultaneously breaches both the ORH and the ORL. Since no tick-by-tick data are used, the chronological order of the two breakout entries cannot be determined. In these cases, we assign the trade direction according to the band nearest to the 5-minute bar's closing price: a long trade is taken if the close is closer to the ORH than to the ORL (with ties assigned long), and a short trade otherwise. This assumption rests on the expectation that the breakout most likely occurred first on the band nearest to the 5-minute bar's closing price, and is designed to preserve the core principle of the ORB framework of taking a single directional trade per session.

Intraday price data are sourced from **IQFeed** (regular trading hours, 1-minute resolution), whereas daily time-series are provided by **Norgate Data**: the full sample covers the period from January 2008 to December 2025.

Our execution and cost assumptions are as follows: fractional share trading is permitted, with commissions of \$0.0035 per share per leg, **doubled on the sell side** to conservatively account for additional sell-side charges; no per-order minimum applies. Market impact is estimated using the I-Star model from Robert Kissell's *The Science of Algorithmic Trading and Portfolio Management*, with parameters $a_1 = 687$, $a_2 = 0.70$, a_3

= 0.72, as calibrated by the author for large-cap stocks. Given that SPY ranks among the most liquid exchange-traded products globally, we believe these parameters offer a conservative estimate of actual market impact.²

² One further refinement concerns the normalization of market impact over time: as portfolio wealth compounds, nominal AUM grows, and the estimated impact cost risks becoming disproportionately large relative to early-period estimates. To prevent this distortion, we rescale market impact at each point in time to correspond to the cost that would be incurred by a fixed base AUM of \$100,000, CPI back-adjusted to maintain real purchasing-power parity throughout the sample: this ensures that slippage estimates remain economically consistent and comparable across the full backtest sample.

Table 1 reports the performance of the **plain-vanilla ORB strategy**. Panel A presents trade-level statistics computed gross of transaction costs and position sizing; Panel B presents portfolio-level statistics fully net of commissions, market impact, and full-notional sizing

Panel A - Trade Statistics								
Hit Ratio (%)	Avg PnL (\$/Sh)	Avg Return (bps)	Active Days (%)	Total Trades	Long Trades	Short Trades	Best Day (bps)	Worst Day (bps)
28.28	0.0336	1.13	100.00	4,529	2,292	2,237	634 (15/10/2008)	-409 (10/10/2008)

Panel B - Portfolio Statistics								
Total Return (%)	CAGR (%)	Volatility (%)	Sharpe (Rf 0%)	Sortino (Rf 0%)	Max Drawdown (%)	Skew	Beta	Alpha (%)
18.7	1.0	9.9	0.14	0.40	40.6	2.66	-0.028	1.75

Table 1: Backtest results for the plain-vanilla 5-minute ORB strategy on SPY (January 2008 – December 2025). α and β are estimated by regressing the strategy's daily returns on SPY

*adjusted daily returns. Statistical significance of the α estimate is denoted by * (p-value < 0.10), ** (p-value < 0.05), and *** (p-value < 0.01).³*

³ Throughout this piece, p-values are used to test the null hypothesis that the relative coefficient equals zero; a statistically significant result indicates that the estimated value is reliably different from zero at the stated confidence level.

The results indicate that, once transaction costs are accounted for, the plain-vanilla strategy delivers essentially flat performance: although the gross trade statistics suggest a positive edge, commissions and market impact erode it almost entirely at the portfolio level.

One statistic in particular deserves attention: the **active days percentage stands at 100%**. This means that on every day in the dataset, the 5-minute opening range was breached in one direction or the other. This outcome is expected: it would be highly unusual for SPY to trade an entire session without ever revisiting the high or low of its opening five minutes, and our ORB framework participates by construction whenever such a breakout occurs.

While unsurprising, this observation highlights a structural limitation of the plain-vanilla approach: **its indiscriminate trading activity**. Participating every day, regardless of conditions, implies a low signal-to-noise ratio and leaves the strategy fully exposed to the cumulative drag of transaction costs.

The baseline results, taken at face value, may appear discouraging. Yet we argue that **the appropriate response is not to discard the framework, but to improve it by reducing trading frequency selectively**, participating only when there are well-founded reasons to expect directional conviction to persist through the session.

Any such enhancements, however, must be grounded in clear economic rationale rather than *ex-post* optimization, so as to ensure genuine predictive content and guard against overfitting.

Enhancing the Base ORB Framework

The Opening Range Direction

In previous work leveraging the 5-minute opening range as a directional signal, particularly *A Profitable Day Trading Strategy for the U.S. Equity Market*, we consistently found value in conditioning trade direction on the opening range body, defined as the open-to-close return of the first 5-minute bar.

The underlying intuition is that the early minutes of the session provide a natural observation window for supply and demand imbalances: if one side of the market holds genuine conviction at the open, its pressure is likely to manifest within this brief initial interval. Accordingly, we would expect the direction of the opening range body to carry information about which side is more likely to prevail through the remainder of the session.

To test this empirically, we take the base strategy described in Section 2 and classify each trade according to whether its direction (i.e. the direction of the first intraday breakout) was aligned or misaligned with the opening range body move.

For each group, we compute the average trade return in basis points, pooled across the full sample: the objective is to assess whether body-breakout alignment carries meaningful predictive content.

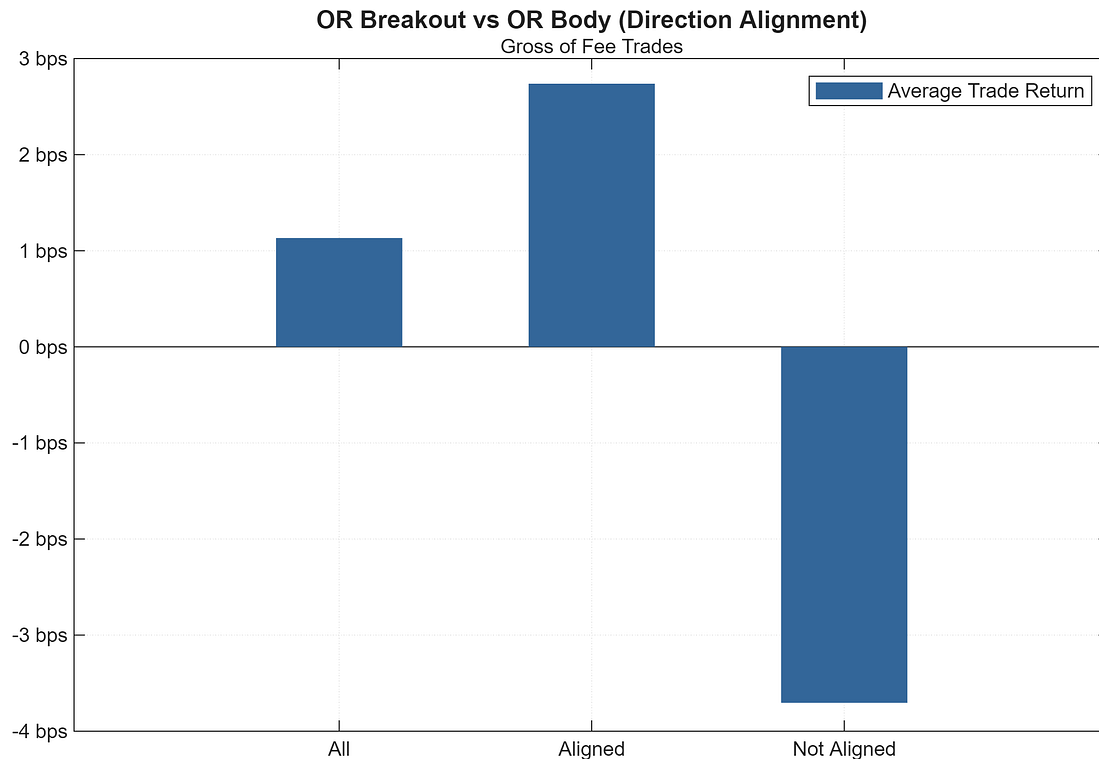


Figure 2: Average trade return in basis points for the plain-vanilla strategy, segmented by whether the breakout direction was aligned or misaligned with the opening range body direction. Results for the full sample are reported under value “All”.

Figure 2 reveals a clear divergence: “Aligned” trades generate average returns **more than double** those of the full unconditional sample, while misaligned trades are unprofitable in aggregate. The implication is clear: conditioning trade initiation on alignment between the breakout direction and the opening range body direction constitutes a clear first improvement.⁴

⁴Sessions in which the open-to-close return of the first 5-minute bar is exactly zero are excluded from this analysis, as the body carries no directional information in those cases.

The Opening Range Magnitude

Having addressed direction, we next consider the magnitude of the opening range body. Even among aligned sessions, it is natural to ask whether a stronger imbalance signal, reflected in a larger body size, leads to better trade outcomes: if the opening range body captures genuine early-session conviction, we would expect trade quality to improve as the body-size threshold is raised.

In particular, we measure body size in relative terms, expressing it as a fraction of the Average True Range with a 14-day lookback period, denoted $ATR(14)$, computed until the prior session's close: we then study how the net-of-fee portfolio total return varies as the minimum body size requirement is progressively increased.

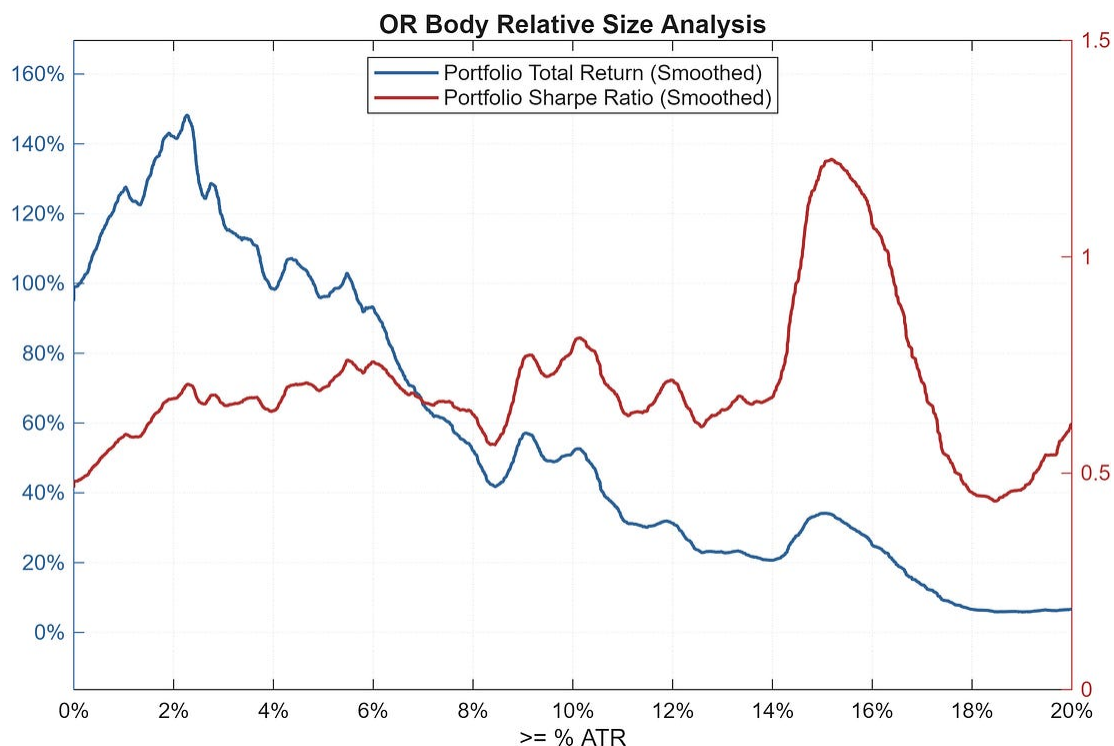


Figure 3: Net-of-fee portfolio total return and Sharpe ratio as a function of the minimum opening range body size, expressed as a percentage of $ATR(14)$. Results are smoothed for clearer visualization.

As shown in Figure 3, portfolio total return rises as the body size threshold increases up to approximately 2% of ATR(14), beyond which it declines. The rising portion reflects improved trade quality, as sessions with a small opening range body display characteristics more consistent with noise than with genuine supply-demand imbalance, and excluding them is beneficial. The declining portion beyond the threshold, however, does not reflect deteriorating trade quality: rather, it reflects a reduction in trade count.

As the red line shows, portfolio Sharpe rises until 4%–5% of ATR(14) and then effectively stabilizes, indicating that stricter OR body size thresholds do not lead to a decline in risk-adjusted performance. In other words, as the minimum body size requirement becomes increasingly stringent, too many sessions are filtered out and the aggregate portfolio return shrinks, even though the remaining individual trades do not lose attractiveness.

Based on these findings, we set the minimum body size threshold at 2% of ATR(14), applied in conjunction with the directional alignment filter introduced above. Layering these two improvements reduces the active trading-days percentage from 100% to approximately **67%**, and raises the net-of-fee CAGR to ~5%. While not exceptional, this represents a meaningful step forward relative to the plain-vanilla baseline and validates the underlying logic of isolating early-session imbalances.

Managing the Trades via Trailing Stops

In prior work, *Volume Weighted Average Price (VWAP): The Holy Grail for Day Trading Systems*, we highlighted the benefits of using the volume-weighted average price (VWAP) as an information-rich reference level for intraday trend-following signals.

Building on those results, we introduce here the concept of an *adjusted VWAP* trailing exit, ported to the current ORB framework. Let $\Delta = \text{ORH} - \text{ORL}$ denote the opening range width: the adjusted VWAP trailing levels for long and short positions are defined as

$$L_t = \max_{\tau \leq t} \max(\text{ORL}, \text{VWAP}_\tau - \Delta)$$

$$S_t = \min_{\tau \leq t} \min(\text{ORH}, \text{VWAP}_\tau + \Delta)$$

During each trading session, a long trade is exited if the 5-minute closing price falls below L_t ; a short trade is exited if it rises above S_t .

As practitioners are aware, VWAP is the most widely adopted intraday execution benchmark among institutional operators and algorithmic trading desks: by construction, it captures the volume-weighted average price of all intraday transactions, making it a natural reference for executing large orders.

For this reason, price exhibits a structural tendency to gravitate around VWAP throughout the session, often revisiting its level as supply and demand find balance (a well-known effect often referred to as VWAP pinning).

Intraday directional programs that exit upon a VWAP breach are particularly exposed to this phenomenon: repeated oscillations around its price level can trigger premature liquidations of otherwise valid trends that would resolve cleanly later in the session.

The adjusted VWAP mechanism addresses this risk directly.

By shifting VWAP downward (for longs) or upward (for shorts) by the full opening range width Δ , we impose a meaningful buffer: price must retrace clearly through the VWAP level before a liquidation event is triggered.

This design seeks to strike a balance between two competing objectives.

On one side, we avoid waiting for a full reversal to the opposite breakout band, which in strongly trending sessions may sacrifice the entire open profit and therefore prove ineffective.

On the other, the Δ buffer absorbs the expected oscillatory behavior around VWAP, substantially reducing the risk of premature exit.

As illustrated in Figure 4, in both representative sessions the trade is carried to the end of the day, an outcome that would not have been possible with a plain VWAP trailing-stop logic.



Figure 4: Illustrative examples of the adjusted VWAP trailing exit: a short ORB trade. A plain VWAP stop would have triggered premature exits, but the Δ buffer effectively absorbs VWAP-pinning dynamics.

As a safeguard mechanism, we impose that the trailing level always respects the opposite band: $L_t \geq ORL$ for longs and $S_t \leq ORH$ for shorts. Allowing the adjusted VWAP to breach the opposite band would

contradict the entry logic itself, since crossing that level is the very condition that triggers trades in the opposite direction; accordingly, whenever the Δ -shifted VWAP drifts beyond this bound, we correct it.

Finally, the expanding maximum in L_t and the expanding minimum in S_t ensure that the trailing level can only advance in the direction of the trade and never relaxes during the session. This monotone property embodies a core principle of trend-following: once a position has moved in our favour, its trailing stop-level should never be pulled back, as discussed in Michael Covel's *Trend Following*.

At the portfolio level, introducing the adjusted VWAP trailing stop raises Sharpe ratio from **0.62 to 0.81**, while reducing maximum drawdown from 17.9% to 14.3%: CAGR also improves, rising from 5.1% to 6.3% (see Table 2 for the full comparison).

Position Sizing

The final enhancement is a volatility-targeting mechanism for position sizing.

Volatility targeting is a well-established methodology in portfolio construction, discussed by Moreira and Muir in *Volatility-Managed Portfolios*, and we adopt here the same implementation developed in *Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)*.

The intuition is straightforward: **rather than always trading at full notional exposure, we scale position size dynamically** so that expected portfolio volatility remains more stable over time.

Concretely, we estimate next-day volatility as the historical volatility of SPY returns over a 14-day lookback period, then size the SPY exposure to target a daily portfolio volatility of 2%, subject to a maximum leverage cap of 4× to comply with standard broker intraday leverage limits⁴.

⁴ The 2% daily volatility target refers to the expected volatility of a passive portfolio that remains continuously invested in SPY throughout the trading day. In our case, however, the same methodology is used only to size an intraday position. Because the strategy is invested for only a fraction of the trading session and has no overnight exposure, its realized daily portfolio volatility is expected to be lower than the 2% target.

Performance Evaluation

Figure 5 shows the cumulative (net-of-fee) equity curve of the fully-enhanced strategy.

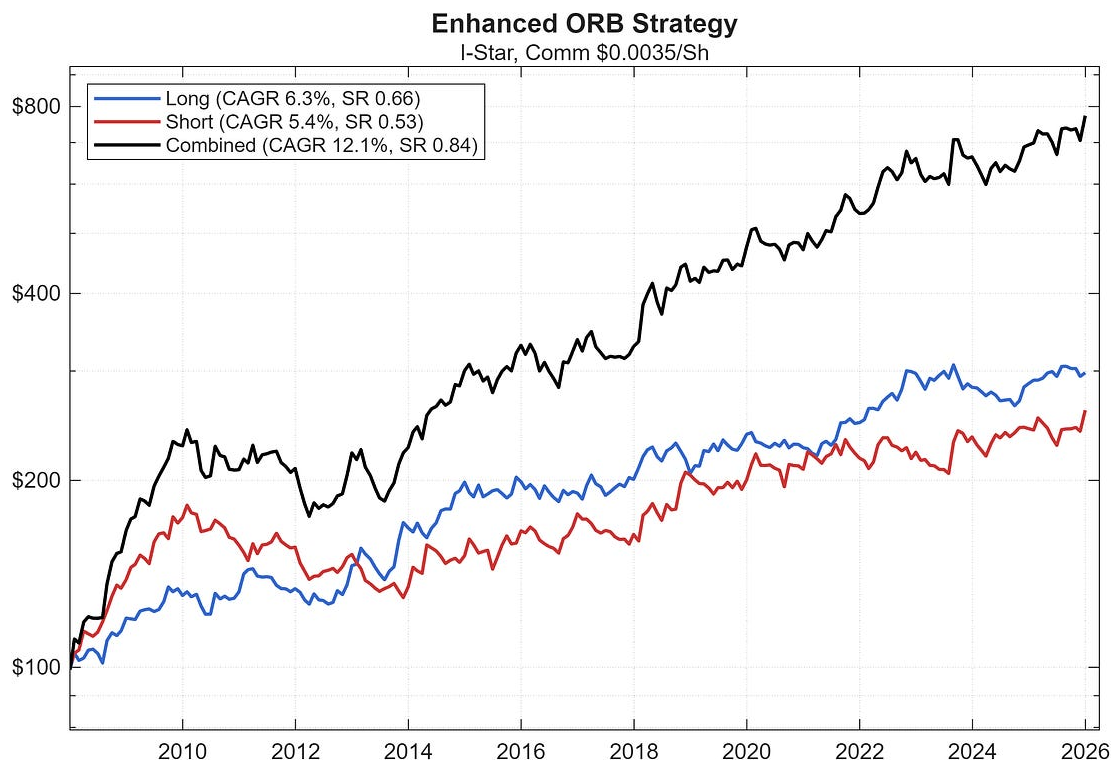


Figure 5: Cumulative net-of-fee portfolio return for the fully-enhanced ORB strategy on SPY, incorporating OR body direction alignment, minimum body size, adjusted VWAP trailing exit, and volatility targeting.

Taken together, the four enhancements introduced in this section demonstrate that the ORB framework, while unremarkable in its plain-vanilla form, can be transformed into a genuinely attractive strategy through a coherent set of principled improvements. As shown in Figure 5, the fully-enhanced configuration achieves net-of-fee cumulative returns well into double-digit territory, with a Sharpe ratio of **0.84**.

The strategy also exhibits a near-zero beta to SPY (slightly negative), a desirable characteristic for active trading approaches, with both the long and short sides of the strategy contributing in similar amounts to the overall PnL. Moreover, the annualized alpha is large (**~12.8%**) and statistically significant at the 1% level: these metrics suggest that the enhanced ORB framework captures returns largely unrelated to, and unexplained by, SPY's pronounced upward drift over the sample period.

Table 2 summarizes the incremental contribution of each previously discussed enhancement, showing how every layer adds measurably to the overall risk-adjusted return profile.

Configuration	CAGR (%)	Volatility (%)	Sharpe (Rf 0%)	Max Drawdown (%)	Skew	Beta	Alpha (%)
Base Model	1.0	9.9	0.14	40.6	2.66	-0.028	1.75
+ OR Body Direction	3.8	8.8	0.47	27.9	3.24	-0.018	4.32**
+ OR Body Relative Size	5.1	8.5	0.62	17.9	3.45	-0.015	5.46***
+ Adjusted VWAP	6.3	7.9	0.81	14.3	3.76	-0.016	6.62***
+ Volatility Targeting	12.1	14.9	0.84	29.8	2.94	-0.026	12.80***

*Table 2: Incremental performance of the ORB strategy as enhancements are added sequentially. Portfolio statistics are net of commissions and market impact. α and β are estimated by regressing the strategy's daily returns on SPY adjusted daily returns. Statistical significance of the α estimate is denoted by * (p -value < 0.10), ** (p -value < 0.05), and *** (p -value < 0.01).*

Conclusion

Our empirical investigation suggests that the long-standing debate surrounding the profitability of Opening Range Breakout (ORB) strategies may have been framed too narrowly.

A naive 5-minute ORB implementation applied to SPY indeed generates little value once realistic commissions and market impact are taken into account.

However, this should not be interpreted as evidence that **the underlying signal has lost its predictive content.**

Instead, our results suggest that **much of the apparent weakness stems from the naive implementation of the ORB framework rather than from the predictive content of the signal itself.**

By introducing **a small set of intuitive, economically motivated enhancements**, we transform the same underlying ORB framework into a strategy delivering a **12.1% net annualized return**, a **Sharpe ratio of 0.84**, and a **12.8% annualized alpha**, while remaining robust under conservative assumptions for commissions and market impact.

More broadly, we believe this case study illustrates an important lesson for systematic trading. **Robust performance often comes not from discovering entirely new trading signals, but from understanding the structural limitations of existing ones** and refining them through disciplined trade selection, thoughtful execution, and sound risk management.

In many cases, the difference between an unremarkable strategy and a compelling one lies not in the signal itself, but in the discipline with which it is implemented.

Naturally, the framework presented here is **not intended to represent the final word on ORB trading.** Several promising avenues remain open

for future research, including the integration of fast alpha signals for execution conditioning, the use of zero-days-to-expiration (ODTE) options as an alternative implementation vehicle, and the construction of diversified cross-asset ORB portfolios.

We hope this work provides a useful foundation for researchers and practitioners interested in developing more robust intraday systematic strategies.

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